

1. Basic Git Commands (For managing repositories)

git init: Initializes a new Git repository in the current directory.

```
chvka@karthik MINGW64 ~/bash-script-project
$ git init
Initialized empty Git repository in C:/Users/chvka/bash-script-project/.git/

chvka@karthik MINGW64 ~/bash-script-project (master)
$ touch karthik

chvka@karthik MINGW64 ~/bash-script-project (master)
$ vi karthik

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git init
Reinitialized existing Git repository in C:/Users/chvka/bash-script-project/.git/

chvka@karthik MINGW64 ~/bash-script-project (master)
$ |
```

git clone <repo_url>: Clones a repository from a remote URL to your local machine.

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git clone https://github.com/chvkarthik-87/script.git
Cloning into 'script'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 1), reused 5 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), done.
Resolving deltas: 100% (1/1), done.
```

git status: Shows the working directory status (which files are modified, staged, etc.).

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    git-clone/
    karthik
    script/

nothing added to commit but untracked files present (use "git add" to track)

chvka@karthik MINGW64 ~/bash-script-project (master)
$
```

git add : Stages a file or files to be committed

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git add g h i j

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   g
        new file:   h
        new file:   i
        new file:   j

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        git-clone/
        karthik/
        script/

chvka@karthik MINGW64 ~/bash-script-project (master)
$
```

git commit -m "message": Commits staged changes to the local repository with a message

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git commit -m "added new files"
[master (root-commit) 40d25cc] added new files
 4 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 g
 create mode 100644 h
 create mode 100644 i
 create mode 100644 j
```

git push: Pushes committed changes to the remote repository.

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 245 bytes | 245.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/chvkarthik-87/script.git
   f5a7dd7..f6b9c5a  master -> master
branch 'master' set up to track 'origin/master'.
```

git pull: Fetches and merges changes from the remote repository to your local working directory.

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git pull --rebase origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 1), reused 5 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 387 bytes | 17.00 KiB/s, done.
From https://github.com/chvkarthik-87/script
 * branch      master      -> FETCH_HEAD
 * [new branch] master      -> origin/master
Successfully rebased and updated refs/heads/master.

chvka@karthik MINGW64 ~/bash-script-project (master)
$ |
```

git fetch: Downloads objects and refs from another repository
(without merging)

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git fetch origin master
From https://github.com/chvkarthik-87/script
 * branch      master      -> FETCH_HEAD
```

git merge : Merges changes from the specified branch into the current branch

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git merge origin/master
Already up to date.
```

git log: Shows the commit history of the repository

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git log
commit f6b9c5a1db8876f9fe821ce882ff1281441d5f70 (HEAD -> master, origin/master)
Author: karthik <chvkarthik07@gmail.com>
Date:   Wed Aug 19 15:03:13 2026 +0530

    added new files

commit f5a7dd72685be315ab5a2348ee46a8370dc92794
Author: karthik <chvkarthik07@gmail.com>
Date:   Wed Aug 19 14:38:02 2026 +0530

    added a file abcd

commit d0ef060b9314fa9ebe5ca16e73fde4fa5ed66153
Author: karthik <chvkarthik07@gmail.com>
Date:   Wed Aug 19 14:00:25 2026 +0530

    adding files
```

git diff: Shows the differences between your working directory and the index (staged changes)

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ echo "new line" >> g

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git diff
warning: in the working copy of 'g', LF will be replaced by CRLF the next time Git touches it
diff --git a/g b/g
index e69de29..86ba82a 100644
--- a/g
+++ b/g
@@ -0,0 +1 @@
+new line

```

2. Branching Commands (Managing branches)

git branch: Lists all branches in your repository.

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch
* master

```

git branch : Creates a new branch.

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git checkout -b karthik
Switched to a new branch 'karthik'

chvka@karthik MINGW64 ~/bash-script-project (karthik)
$ git branch
* karthik
  master

chvka@karthik MINGW64 ~/bash-script-project (karthik)
$

```

git checkout <branch_name>: Switches to an existing branch.

```

chvka@karthik MINGW64 ~/bash-script-project (karthik)
$ git checkout master
Switched to branch 'master'
M      g
Your branch is up to date with 'origin/master'.

chvka@karthik MINGW64 ~/bash-script-project (master)
$ |

```

git checkout -b <branch_name>: Creates and switches to a new branch in one step

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git checkout -b vinni
Switched to a new branch 'vinni'

chvka@karthik MINGW64 ~/bash-script-project (vinni)
$ |

```

git merge <branch_name>: Merges changes from the specified branch into the current branch.

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git merge master
Already up to date.
```

git rebase <branch_name>: Applies changes from one branch onto another branch (instead of merging)

```
$ git restore .

chvka@karthik MINGW64 ~/bash-script-project (karthik)
$ git rebase master
Current branch karthik is up to date.

chvka@karthik MINGW64 ~/bash-script-project (karthik)
$ git status
On branch karthik
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        git-clone/
        karthik/
        script/

nothing added to commit but untracked files present (use "git add" to track)
```

git branch -d <branch_name>: Deletes the specified branch (locally)

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch -d vinni
Deleted branch vinni (was f6b9c5a).

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch
  karthik
* master
```

git branch -D <branch_name>: Forcefully deletes a branch (locally).

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch -D ganesh
Deleted branch ganesh (was f6b9c5a).

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch
  karthik
* master

chvka@karthik MINGW64 ~/bash-script-project (master)
$
```

git remote set-head origin <branch_name>: Sets the default branch for the remote repository.

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git remote set-head origin master

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git remote show origin
* remote origin
  Fetch URL: https://github.com/chvkarthik-87/script.git
  Push URL: https://github.com/chvkarthik-87/script.git
  HEAD branch: master
  Remote branch:
    master tracked
  Local branch configured for 'git pull':
    master merges with remote master
  Local ref configured for 'git push':
    master pushes to master (up to date)

```

3.Remote Commands (Managing remotes and syncing with the cloud)

git remote -v: Shows the remote repository URLs

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git remote -v
origin https://github.com/chvkarthik-87/script.git (fetch)
origin https://github.com/chvkarthik-87/script.git (push)

```

git remote add : Adds a new remote repository.

```

chvka@karthik MINGW64 ~/d/gitpractice (master)
$ git remote add origin https://github.com/chvkarthik-87/script.git

```

git remote remove : Removes a remote repository

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git remote remove origin

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git remote -v

chvka@karthik MINGW64 ~/bash-script-project (master)
$

```

git push origin <branch_name>: Pushes a branch to a remote repository.

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git push origin master
Everything up-to-date

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    git-clone/
    karthik/
    script/

nothing added to commit but untracked files present (use "git add" to track)

```

git push -u origin <branch_name>: Pushes a branch to a remote repository and sets the upstream for the branch.

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git push -u origin master
Everything up-to-date
branch 'master' set up to track 'origin/master'.

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git push
Everything up-to-date

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch
  karthik
* master

```

git pull origin <branch_name>: Fetches and merges a branch from the remote repository

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git pull origin master
From https://github.com/chvkarthik-87/script
* branch      master      -> FETCH_HEAD
Already up to date.

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git branch
  karthik
* master

```

git fetch origin: Downloads objects and refs from the remote repository.

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git fetch origin

```

The output did not appear because there were no new changes in the remote GitHub repository that your local repository needed to fetch.

git remote show origin: Displays detailed information about a remote repository

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git remote show origin
* remote origin
  Fetch URL: https://github.com/chvkarthik-87/script.git
  Push URL: https://github.com/chvkarthik-87/script.git
  HEAD branch: master
  Remote branch:
    master tracked
  Local branch configured for 'git pull':
    master merges with remote master
  Local ref configured for 'git push':
    master pushes to master (up to date)

```

4. Staging and Committing (Managing changes)

git add . : Stages all modified files in the working directory

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git add .
warning: in the working copy of 'karthik', LF will be replaced by CRLF the next time Git touches it
warning: adding embedded git repository: script
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> script
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached script
hint:
hint: See "git help submodule" for more information.

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   karthik
        new file:   script

```

git add -A: Stages all modified files (including file deletions).

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git add -A

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   karthik
        new file:   script

```

git reset : Unstages a file (reverts it to the working directory).

```

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git reset

chvka@karthik MINGW64 ~/bash-script-project (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        karthik
        script/

nothing added to commit but untracked files present (use "git add" to track)

```

git commit --amend: Modifies the last commit (can change the commit message or add changes)

```
chvka@karthik MINGW64 ~/bash-script-project (master)
$ git commit --amend -m "added new file"
[master 423d34d] added new file
Date: Wed Aug 19 15:03:13 2026 +0530
4 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 g
create mode 100644 h
create mode 100644 i
create mode 100644 j
```